Ashkan Pirmani

Applied Machine Learning Scientist

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Summary

Applied machine learning scientist with seven years on real-world health data that cannot be pooled, from data access and pipelines to models and publication. Built the federated route that added 3,527 records, 45.5% more, to a worldwide study of multiple sclerosis medicines and COVID-19, whose findings informed the global COVID-19 advice for people with multiple sclerosis (JMIR Medical Informatics, 2023). Personalizing a federated model to each of 32 countries improved AUC-PR by 31% over the same model without personalization, on 26,246 patients (npj Digital Medicine, 2025). Now developing propensity score methods for drug safety studies across small data sources, under an Innovation Mandate with Johnson & Johnson Innovative Medicine as industry partner.

Experience

Applied Machine Learning Scientist, Drug Safety Methods (Innovation Mandate)

Hasselt University, Belgium | Sep 2026 to Present

- Won a personal Innovation Mandate grant from the Flemish government's innovation agency (VLAIO), with Johnson & Johnson Innovative Medicine as industry partner.
- Developing personalized federated learning methods that keep treatment groups comparable in propensity score analyses when each data source has few patients or the treatment is new.
- Validating them first on an international multiple sclerosis network against criteria set in advance, then on Optum US health records and insurance claims.

Applied Machine Learning Scientist (Postdoctoral Researcher)

KU Leuven and Hasselt University, jointly funded, Belgium | Dec 2024 to Aug 2026

- Personalizing a federated model to each of 32 countries improved ROC-AUC by 7.2% and AUC-PR by 31% over the same model without personalization, and beat pooling all records, in predicting confirmed disability worsening within two years for 26,246 patients. Designed AdaptiveDualBranchNet for it. First author, npj Digital Medicine (2025).
- FLoRank, a personalized federated method built on adaptive low-rank factorization, improved AUC-PR by 24% over standard federated learning (FedAvg) and F1 by 8% over LoRA, on registry data from over 26,000 patients at 146 centers. Prototype. Awarded an ECTRIMS grant for scientific quality (ECTRIMS 2025).
- FLkit: led the design of an open guide for teams starting a study across centers, with 29 contributors, supported by ELIXIR Belgium. First author, arXiv (2026).
- Identified five barriers that stop research software from being reused (scope, workflow, ecosystem maturity, infrastructure mandates and institutional trust), by retiring my own framework, FL4E, when the next study needed a different approach. First author, Medical Informatics Europe (2026).
- Organizing team of OHDSI Europe 2025. Invited talks at Roche (federated learning team for multiple sclerosis), Flower Monthly and the North American Society of Artificial Intelligence in Multiple Sclerosis.
- Coordinated with data holders, clinicians and legal teams in more than 30 countries. Supervised seven master's theses in artificial intelligence since 2021.

Machine Learning Researcher (Dual PhD)

KU Leuven and Hasselt University, Belgium | Sep 2019 to Nov 2024

- Built a federated route for four registries that could not send patient records to a worldwide study of multiple sclerosis medicines and COVID-19: a container at each registry turned the records into grouped counts, and only the counts left. It added 3,527 records, 45.5% more, and the study's findings informed the global COVID-19 advice for people with multiple sclerosis. First author, JMIR Medical Informatics (2023).
- Designed the study's data pipeline: one core data set agreed across 18 registries, three routes for data to come in, and the same quality checks on every stream.
- In over 5,500 people with multiple sclerosis, the anti-CD20 medicines ocrelizumab and rituximab were associated with more severe COVID-19 than glatiramer acetate, and interferon-β was not protective compared with other medicines. My part was data acquisition, the pipeline and the federated route. Shared first author, Neurology: Neuroimmunology & Neuroinflammation (2022) and Multiple Sclerosis and Related Disorders (2022).
- Led the disability progression study from the data access proposal to MSBase and the ethics approval to the experiments. AdaptiveDualBranchNet shares part of the model across countries and keeps part local. Built in Python, PyTorch, Flower and Ray on the Flemish Supercomputer Center's cluster.

- ATHENA, a public-private partnership with Janssen Pharmaceutica, funded part of my PhD. Worked on the analysis in its machine learning work package.
- Built FL4E, an open source framework where each center decides whether its records leave. On a public classification benchmark, letting half the centers send their records kept 97.5% of the fully federated score. First author, JMIR Formative Research (2024).
- Taught in Data Science in Healthcare, Healthcare Engineering program, Hasselt University (2022 to 2024).

Research Assistant, Decision Support Systems Lab
Iranian Research Institute for Information Science and Technology (IranDoc), Tehran, Iran | Feb 2017 to Mar 2019

- Built a hybrid simulation of Iran Post's national parcel network (system dynamics and agent-based modeling in R and NetLogo) and searched it with a multi-objective genetic algorithm (NSGA-II). Best hub layout: 15% lower projected transport cost, in simulation. Delivered as a decision support dashboard for executives.

Education

Doctor of Engineering Science (PhD), Electrical Engineering, KU Leuven, Belgium | Nov 2024

Doctor of Philosophy (PhD), Biomedical Sciences, Hasselt University, Belgium | Nov 2024

Dual PhD, fully funded jointly by both universities. Thesis: From Centralized to Federated: The Journey of Data in Healthcare. Supervisors: Yves Moreau and Liesbet M. Peeters. Co-supervisor: Niels Hellings.

MSc, Industrial Engineering (Socio-Economic Systems), Kharazmi University, Tehran, Iran | Jan 2019

MBA, Quality Engineering, Tose'e Institute, Tehran, Iran | 2017

Skills

Methods: machine learning, neural networks, federated learning, personalized federated learning, federated analysis, privacy-preserving machine learning, predictive modeling, survival analysis, classification on clinical data, model evaluation for imbalanced outcomes (ROC-AUC, AUC-PR).

Data: real-world data (RWD), real-world evidence (RWE), observational data, disease registries, cohort building and phenotype definition, registry harmonization, data quality, data governance, data feasibility, OMOP common data model (OHDSI), FAIR data.

Tools: Python, PyTorch, Flower, Ray, scikit-learn, pandas, R, SQL, Docker, Kubernetes, continuous integration, high performance computing.

Collaboration: grant writing, project leadership, work with clinicians, data holders and legal teams, mentoring.

Awards

- Innovation Mandate, Flanders Innovation & Entrepreneurship (VLAIO), 2026. Industry partner: Johnson & Johnson Innovative Medicine.
- ECTRIMS grant for scientific quality, 2025, for the FLoRank abstract.
- Fully funded PhD scholarship, KU Leuven and Hasselt University, 2019 to 2024.

Selected Publications

- Pirmani A, et al. Personalized federated learning for predicting disability progression in multiple sclerosis using real-world routine clinical data. npj Digital Medicine, 2025.
- Pirmani A, et al. The Journey of Data Within a Global Data Sharing Initiative: A Federated 3-Layer Data Analysis Pipeline to Scale Up Multiple Sclerosis Research. JMIR Medical Informatics, 2023.
- Simpson-Yap S, Pirmani A (shared first authors), et al. Updated results of the COVID-19 in MS global data sharing initiative: anti-CD20 and other risk factors associated with COVID-19 severity. Neurology: Neuroimmunology & Neuroinflammation, 2022.
- Simpson-Yap S, Pirmani A (shared first authors), et al. Severity of COVID19 infection among patients with multiple sclerosis treated with interferon-β. Multiple Sclerosis and Related Disorders, 2022.
- Pirmani A, et al. Accessible Ecosystem for Clinical Research (Federated Learning for Everyone): Development and Usability Study. JMIR Formative Research, 2024.
- Pirmani A, Moreau Y, Peeters LM. Good for All, Not Good Enough for One: Reuse Dilemma in Federated Learning. Studies in Health Technology and Informatics (Medical Informatics Europe), 2026.
- Pirmani A, Vermeulen I, et al. Development and Design of FLKit: A Structured Onboarding Toolkit for Federated Learning in Health and Life Sciences. arXiv, 2026.

Full list: ashkan-pirmani.github.io/publications.html

Languages

Azerbaijani (native), Persian (native), English (full professional), Turkish (full professional), Dutch (limited working).